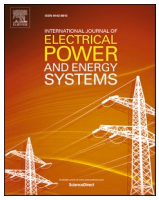
Electrical Power and Energy Systems 155 (2024) 109582



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes





Voltage stability improvement in distribution networks by using soft open points

Kambiz Mardanimajd [a], Shahram Karimi [a,*], Amjad Anvari-Moghaddam [b]

[a] Department of Electrical Engineering, Razi University, Kermanshah, Iran
[b] Department of Energy (AAU Energy), Aalborg University, 9220 Aalborg, Denmark

ARTICLE INFO

Keywords:
Voltage stability index
Soft open point
Renewable energies
Active distribution network

ABSTRACT

The integration of renewable energies is gaining popularity as a result of their sustainability and green nature. On the other hand, aged power infrastructure not only has various issues in accommodating such resources, but also poses risks to the distribution network operation. To this aim, soft open points (SOPs) can be used for supporting distribution networks. Although SOP has been used for different purposes, the potential of SOP for increasing the voltage stability entails further study. As a result, an efficient strategy for the operation of SOPs in distribution networks is developed in this study, which considers the voltage stability index as an objective function. The developed framework is optimized using the particle swarm optimization method, which takes network, resource, and other technical limitations into account. The uncertainties of loads and renewable energies are also modeled through the scenario-based method. The suggested approach is tested on an IEEE standard distribution network under two conditions, comprising the normal and fault situations. After analyzing the various case studies and experiments, the effectiveness and applicability are verified accordingly. The findings show that SOP has a significant impact on improving voltage stability of distribution networks, throughout the normal and fault conditions.

1. Introduction

The use of fossil fuels to generate electricity has long been the foundation of our worldwide energy systems, propelling economies and supporting our contemporary way of life. This increasing reliance on fuel-based power systems, however, has come at a high cost, since it has played a key part in aggravating the severe threat of climate change. Our reliance on fossil fuels has resulted in the release of astonishing volumes of glasshouse gases into the atmosphere, resulting in a major shift in Earth's climate patterns and a slew of severe environmental impacts [1,2]. Furthermore, the influence of climate change caused by fuel-based power systems goes beyond temperature fluctuations. It has serious implications for biodiversity, ecosystems, and human livelihoods. These consequences, ranging from coral bleaching jeopardizing marine ecosystems to droughts impacting agricultural production, highlight the necessity of shifting to sustainable and climate-resilient energy sources [3,4]. Despite the mentioned difficulties, there is a chance to reinvent our energy future. We open the door to investigating revolutionary alternatives by understanding the harmful link between fuel-based power systems and climate change [5]. Distributed generation (DG) resources, such as solar, wind and electric vehicle aggregations [6] offer a no-carbon, sustainable, and ecologically responsible alternative. Adopting clean energy solutions not only reduces glasshouse gas emissions, but also promotes economic growth, and energy security [7,8].

The usage of distributed energy resources has grown in response to techno-economic issues such as limitations of distribution line capacities, as well as environmental aspects [9]. Generally speaking, DG sources are small-scale power generating units that are almost allocated in the distribution side [10]. There are different types of DGs which most popular/common ones include wind turbine, and solar panels. Due to their environmentally beneficial properties and sustainability, wind turbines and solar panels are being used around the world more and more [11]. Although such power sources bring significant advantages for both operators and societies, their generation profiles completely depend on meteorological parameters. Accordingly, existing distribution networks may encounter voltage violations under pierce of such resources.

With increasing the power requested by customers, efforts have been made to evaluate the ability of distribution systems to meet the growth of electricity demand with the existing power systems. In order to secure operation of distribution networks, different strategies have been

* Corresponding author.

https://doi.org/10.1016/j.ijepes.2023.109582
Received 23 April 2023; Received in revised form 8 August 2023; Accepted 13 October 2023
Available online 18 October 2023
0142-0615/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

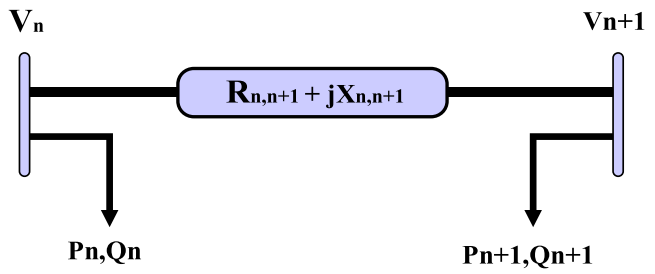


Fig. 1. A sample two-bus system.

introduced so far. An optimization model based on non-dominated sorting genetic algorithm was provided in [12] for improving the efficiency of the system by minimizing power loss, as well as enhancing the voltage stability and security. Similarly, authors in [13] used particle swam optimization algorithm to allocate renewable DGs in distribution systems with the aim of voltage stability improvement and real power loss minimization. Reference [14] proposed a planning approach for static var compensator allocation in the presence of online tap changer transformers and generators or synchronous condensers to enhance the stability of the systems. Researchers in [15] allocated different types of DGs in distribution networks for improving stability enhancement of distribution networks. Distribution static compensator (DSTATCOM) was another equipment which was placed for voltage stability improvement of distribution systems [16].

Generally, soft open point (SOP) is a power electronic-based equipment, which is designed based on back-to-back voltage source converters (VSCs) [17] to be used in distribution networks. In more details, distribution networks have radial structure and the buses are connected by sectionalizing switches. Apart from these switches, there are tie-line switches which are open normally [18]. The SOP is installed in the place of tie lines and therefore connects two adjacent feeders [19]. When connecting two adjacent feeders, operators are able to control the power flow among these feeders which brings various advantages such as voltage regulation and grid integration of more renewables. Regarding SOP operating in distribution networks, some investigations have been done by researchers. As an example, authors of [20] proposed an operational scheme for SOP in distribution systems for optimizing two objective functions, comprising real power loss minimization and feeder load balancing. From this research, it was clear that SOP can play a significant role in power loss minimization and load balancing. On the same track, in [21], researchers had SOP operated for optimizing three

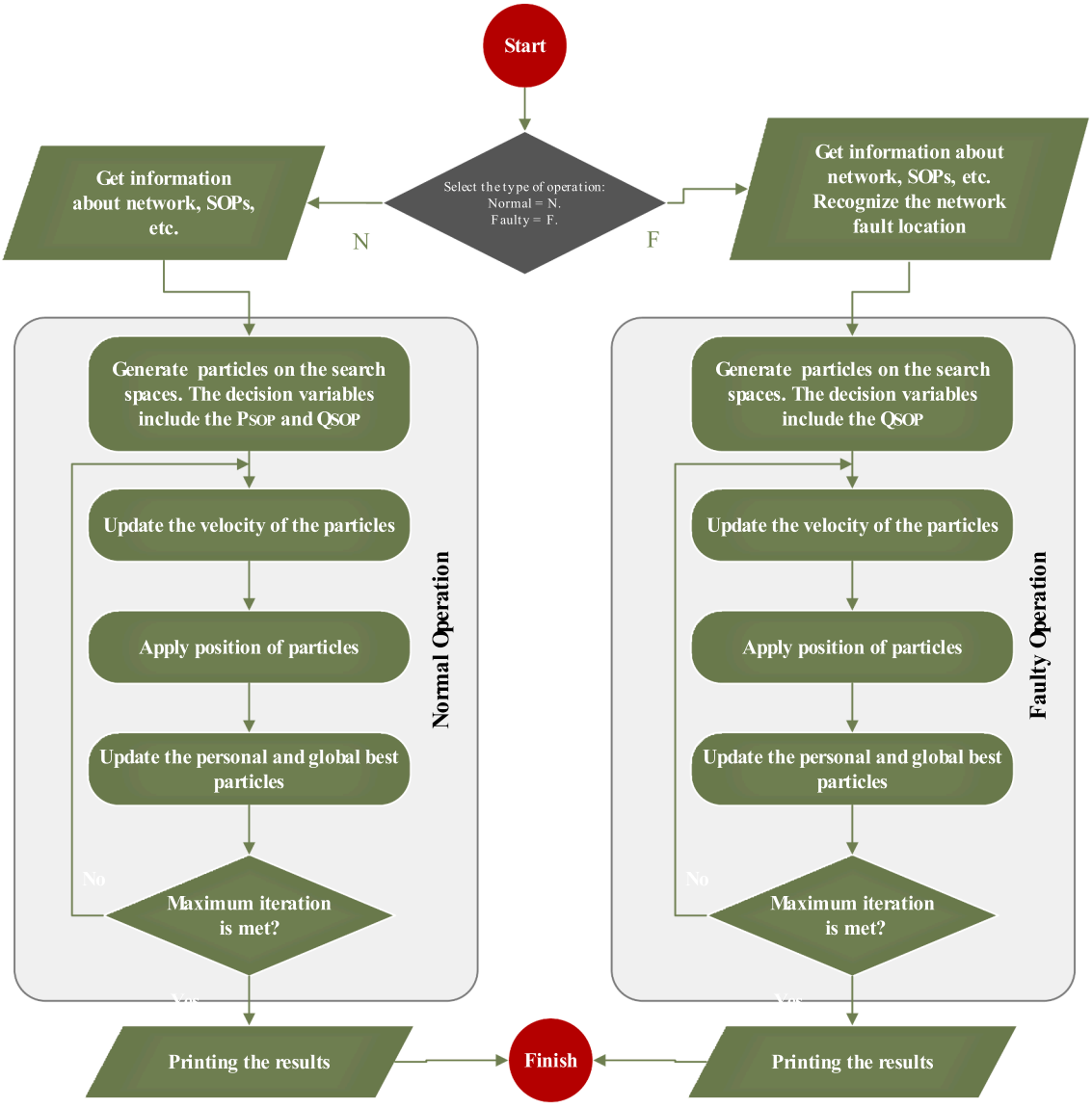


Fig. 2. The flowchart of the proposed approach.

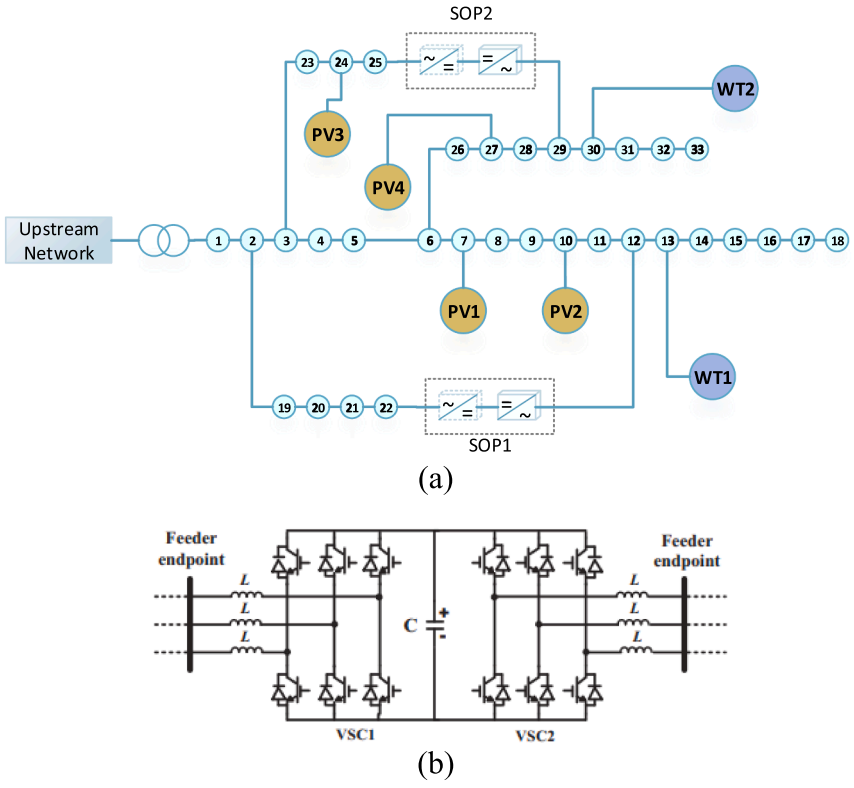


Fig. 3. (a) The schematic of under study network (b) Structure of SOP.

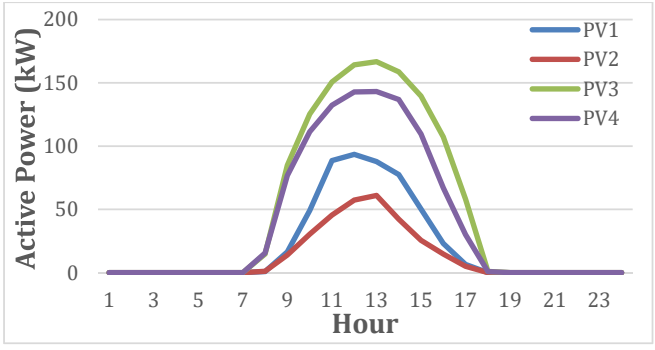


Fig. 4. The forecasted values of PVs' generation.

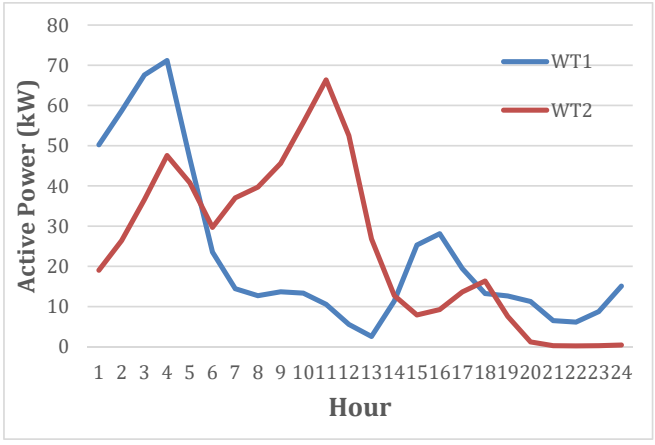


Fig. 5. The forecasted values of WTs' generation.

functions: voltage profile, line load balancing, and energy loss minimization. Higher hosting capacities of distribution generations in distribution systems by SOP was investigated in [22]. Load balancing via SOP was also devised in [23]. Authors in [24] analyzed the potential of SOP in reliability enhancement of distribution networks. Voltage and reactive power were controlled in distribution systems by SOP in [25]. Simultaneous management of SOP and online tap changer transformer was investigated for voltage deviation minimization [26]. Congestion of lines has also been relieved by SOP in [27]. Current power grids are categorized into several microgrids with the aim of decreasing the reliability risk as well as enhancing the restoration. As an example, Authors in [28] developed an optimization approach for effectively partitioning distribution networks via the SOPs. From the results, it could be seen that the system operational cost as well as technical parts has improved significantly. Besides, in [29] assessed the potential of SOPs on system restoration of power grids. As can be observed through the results, SOPs are viable strategy to restore the loads and resilience enhancement of the grid. From this review, it can be concluded that SOP has been a great equipment to improve the performance of the system such as power loss minimization, reliability enhancement, load balancing, etc. However, utilizing SOPs to improve the voltage stability of distribution systems has not been investigated adequately.

To bridge the aforementioned gaps, this paper introduces an optimization approach based on particle swarm optimization (PSO) algorithm to optimally manage SOPs in the distribution networks with the aim of voltage stability improvement. In more detail, because of being on the mitigation of climate change, renewable resources integration has been one of the influential alternatives in power system sector. However, the current network suffers from hosting significant renewable resources because of the aged infrastructure of power systems and stability issues. One of the most important problems is the voltage stability issues as a result of the unpredictability of renewable generations and the occurrence of faults. To this issue end, firstly, a new voltage stability factor is utilized as an objective function and then the problem

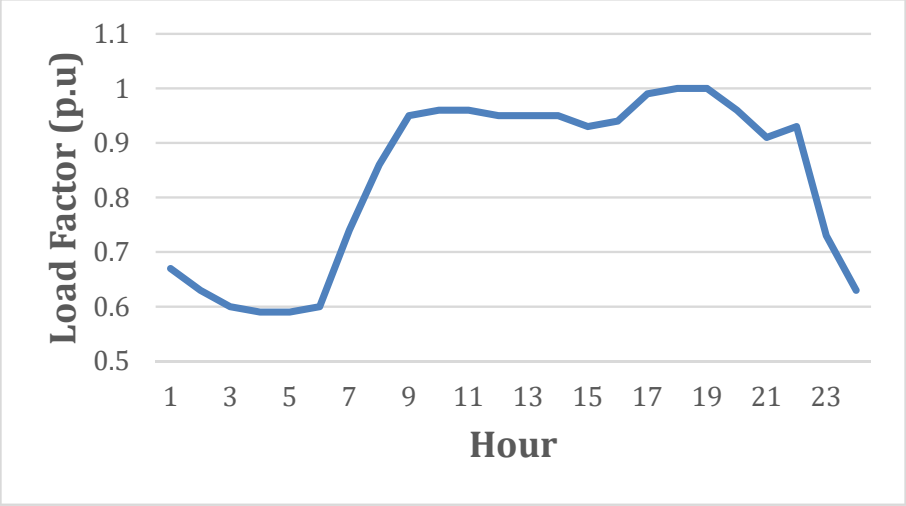


Fig. 6. The load factor of the network.

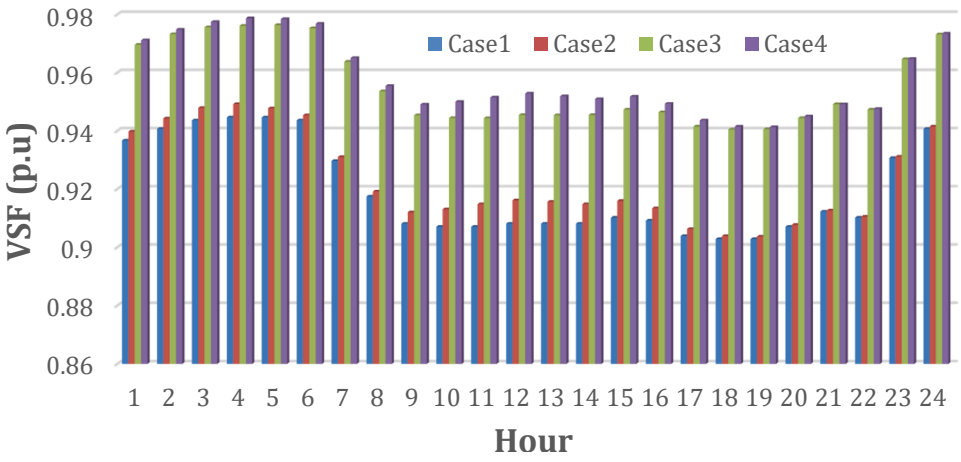


Fig. 7. The minimum VSF over the period.

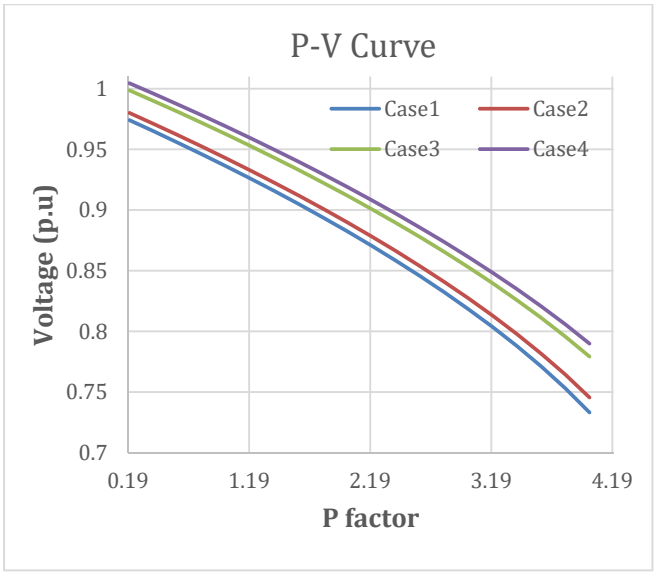


Fig. 8. The P-V curve for different case studies.

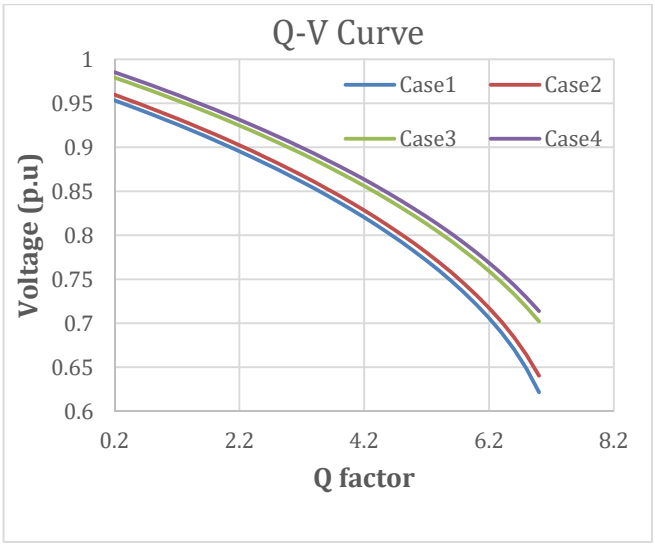


Fig. 9. The Q-V curve for different case studies.

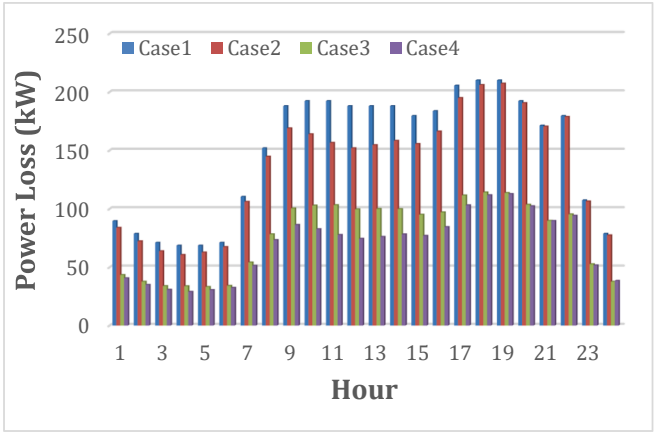


Fig. 10. The power loss of the network.

is optimized subject to different constraints related to the distribution network, SOP, etc. The capability of the developed framework is then demonstrated from implementing it on a typical 33-bus distribution system under two conditions. In other words, the network is firstly analyzed when it is operated under normal condition to recognize the potential of SOPs to deal with voltage stability issues. In the next step, the condition of network by performing a fault is made worse to check the possibility of SOPs to cope with the incident, resulting in not collapsing voltage in the network. As a whole, the novelty aspects and contributions of this study can be summarized as follows:

- Assessing the effect of SOP on voltage stability of distribution networks under the penetration of solar and wind energies.
- The loadability of network integrated with SOPs is evaluated by means of the P-V and Q-V curves which lead to recognize the stability margin of the network.
- The potential of SOP in improving the voltage stability of network under the fault and unbalanced network conditions is also evaluated.
- The effect of uncertain parameters, load and renewable energy resources, is assessed on the model by scenario-based method.

The rest of paper is organized as follows. In section 2, the problem formulation is discussed. Section 3 introduces the PSO algorithm. The potential of SOP for stability improvement of the distribution networks is discussed by taking into account different case studies in section 4. Finally, the conclusion and future works are provided in section 5.

2. Problem definition

2.1. Objective function

Voltage stability has become one of the most important aspects in operation of power systems, particularly in distribution networks which bring the power down to the consumers and accommodate renewable resources. In order to measure voltage stability, several methods such as nose curve technique, sensitivity analysis, voltage stability factors (VSFs) have been proposed so far from which the VSF is deemed as a faster and promising tool for stability analysis of distribution networks. Here, a stability factor which has been introduced in [30] is utilized to assess the impact of SOP on stability improvement.

Let's consider Fig. 1 as a first and fundamental step to easily comprehend voltage stability factor.

As can be seen from this two-bus sample network, the current ($I^2_{n,n+1}$) flows between these buses as given in the following equation.

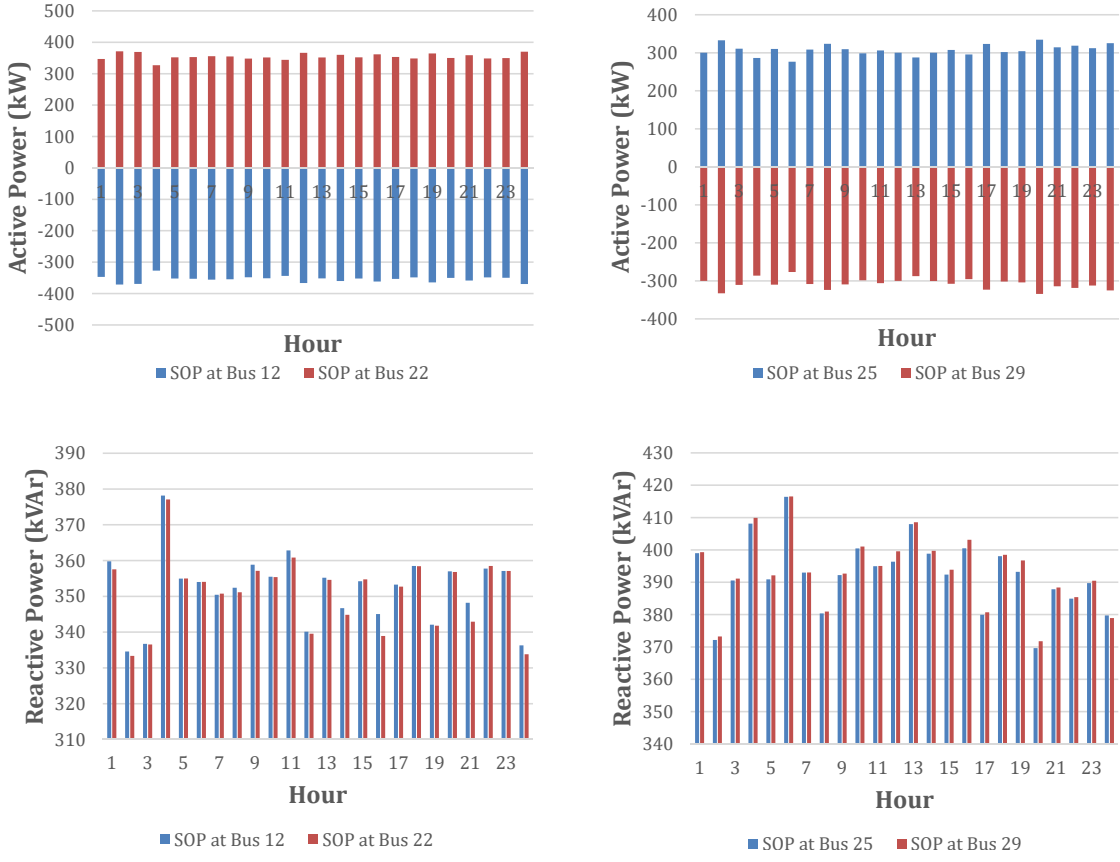


Fig. 11. The amount of powers provided by SOPs in case 3.

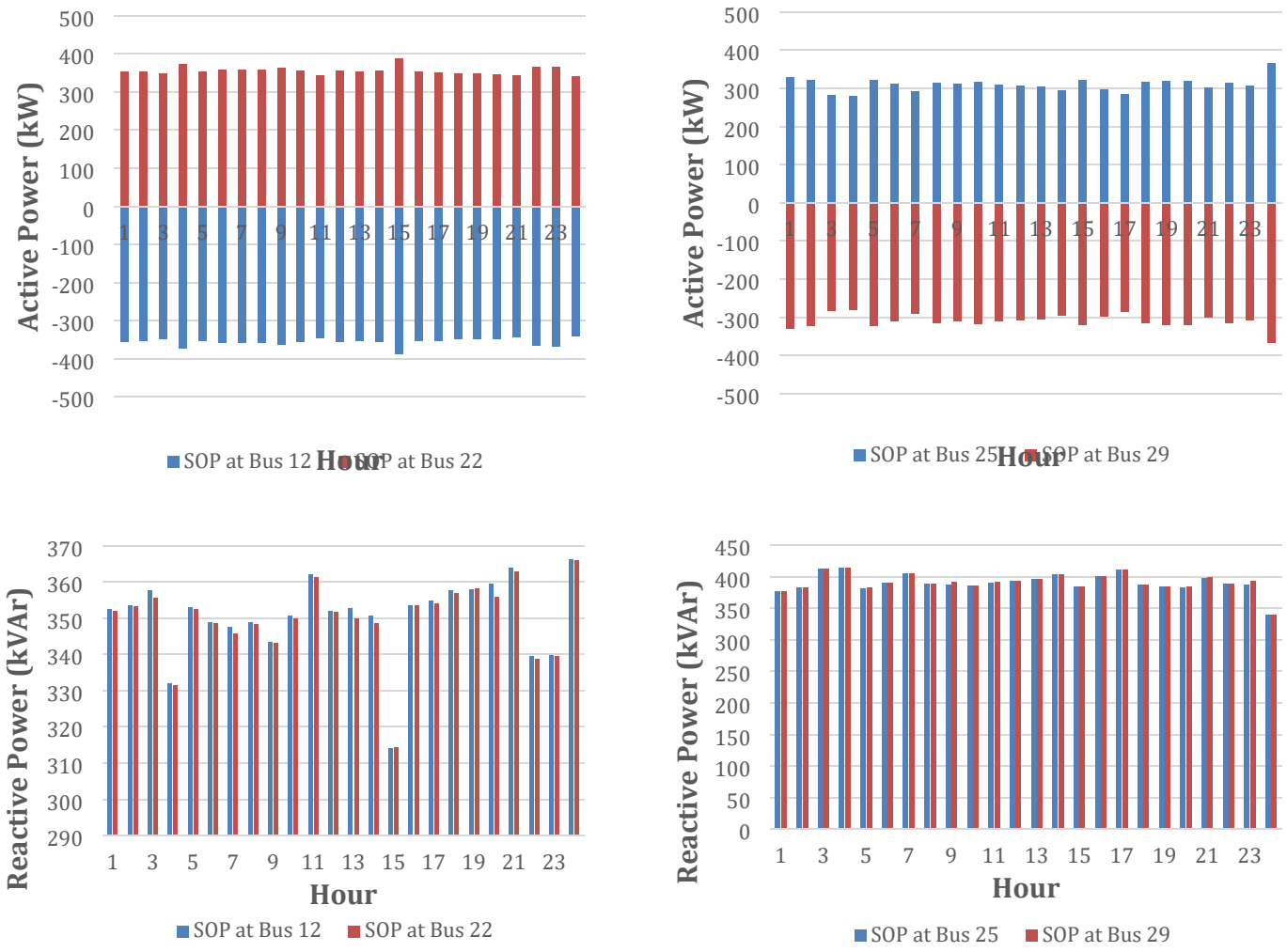


Fig. 12. The amount of powers provided by SOPs in case 4.

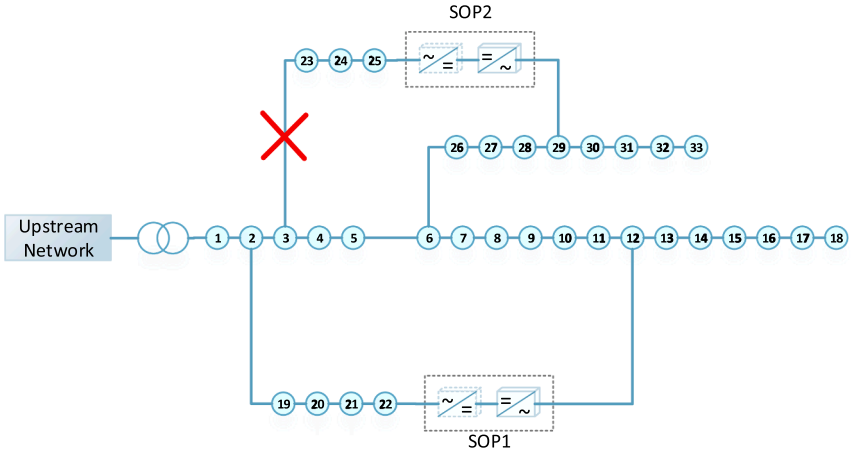


Fig. 13. The 33-bus test network under fault situation.

$$I^2_{n,n+1} = \left(\frac{P^2_{n+1} + Q^2_{n+1}}{V^2_{n+1}}\right) \quad (1)$$

P_{n+1}, Q_{n+1} and V_{n+1} in the mentioned expression denote the active and reactive powers and voltage at bus $n+1$, respectively.

The active and reactive power losses of the shown sample network are calculated as follows:

$$Ploss_{n,n+1} = R_{n,n+1}\left(\frac{P^2_{n+1} + Q^2_{n+1}}{V^2_{n+1}}\right) \quad (2)$$

$$Qloss_{n,n+1} = X_{n,n+1}\left(\frac{P^2_{n+1} + Q^2_{n+1}}{V^2_{n+1}}\right) \quad (3)$$

where, resistance and reactance of the line were respectively

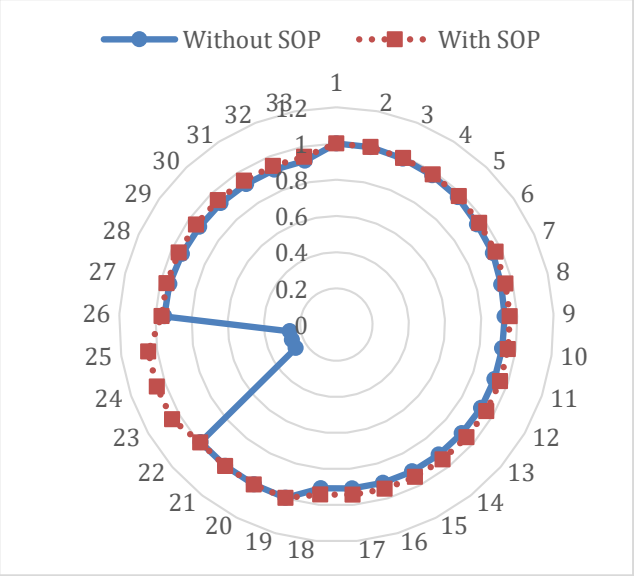


Fig. 14. Voltage profile of balanced network under fault condition.

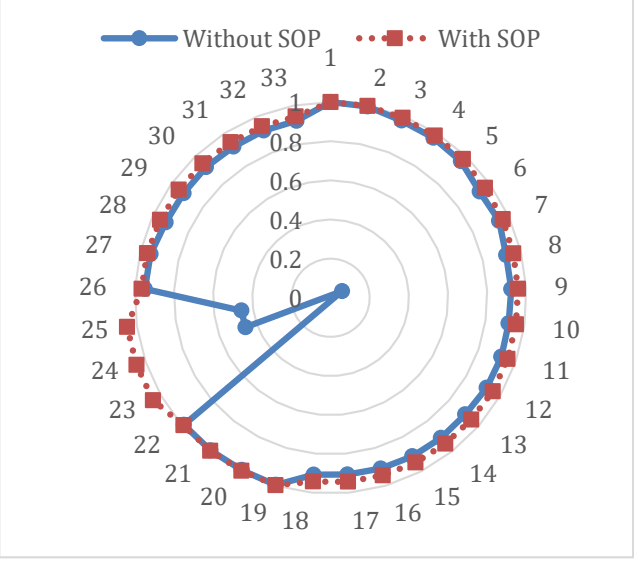


Fig.15. VSF of balanced network under fault condition.

indicated by $R_{n,n+1}$ and $X_{n,n+1}$.

The current transferred from n to $n+1$ can also be expressed as follows:

$$I^2_{n,n+1} = \left(\frac{Ploss^2_{n,n+1} + Qloss^2_{n,n+1}}{(V_n - V_{n+1})^2}\right) \tag{4}$$

In the above expression, $I^2_{n,n+1}$ is substituted by (1) and thus the following equation is obtained:

$$\frac{P^2_{n+1} + Q^2_{n+1}}{V^2_{n+1}} = \frac{Ploss^2_{n,n+1} + Qloss^2_{n,n+1}}{(V_n - V_{n+1})^2} \tag{5}$$

After replacing the relations of $Ploss_{n,n+1}$ and $Qloss_{n,n+1}$ in above expression, the following equation is obtained:

$$\left(P^2_{n+1} + Q^2_{n+1}\right).\left(R^2_{n,n+1} + X^2_{n,n+1}\right) = \left(V^2_{n+1} - V_{n+1}.V_n\right)^2 \tag{6}$$

If the positive root of the above expression is taken, the following equation is obtained:

$$S_{n+1} = \frac{(V^2_{n+1} - V_{n+1}.V_n)}{Z_{n,n+1}} \tag{7}$$

The apparent power at bus $n+1$ is signified by S_{n+1}; the amount of line impedance is also shown by.$Z_{n,n+1}$

For critical power flowing at $n+1$, the following expression is obtained:

$$\frac{dS_{n+1}}{dV_{n+1}} = \frac{(2V_{n+1} - V_n)}{Z_{n,n+1}} = 0 \tag{8}$$

If system is operated under stable condition, the following expression is obtained.

$$\frac{dS_{n+1}}{dV_{n+1}} > 0 i.e. (2V_{n+1} - V_n) > 0 \tag{9}$$

Finally, the amount of voltage stability factor at the bus $n+1$ is calculated as:

$$VSF_{n+1} = (2V_{n+1} - V_n) \tag{10}$$

In order to extend the above voltage stability factor for larger-scale networks, the following relation is used.

$$VSF_T = \sum_n^{Nbr} (2V_{n+1} - V_n) \tag{11}$$

here,VSF_T shows the summation of VSF over the entire network; *Nbr* signifies the number of branches.

Finally, operators who tend to have stable system must increase the VSF. This could be considered as objective function as given in (12):

$$\text{Obj_Fun} = min\left(1/\sum_n^{Nbr} (2V_{n+1} - V_n)\right) \tag{12}$$

2.2. Constraints

The examined optimization problem in this study includes some equality and inequality constraints. Some of these constraints are associated with distribution networks such as the voltage boundaries, maximum capacity of branches, and power balances that are expressed as follows [6,31–33]:

- Acceptable voltages and capacity of branches

$$V_i^{min} \le V_i \le V_i^{max} \tag{13}$$

$$I_j \le I_j^{max} \tag{14}$$

V_i and I_j in the above expressions respectively demonstrates the bus voltage and branch current of network; V_i^{min}, V_i^{max} and I_j^{max} denote the boundaries of bus voltage and branch current, respectively.

- The demand–supply balance

$$P_l^{inj} = \sum_{k=1}^{N_{bs}} V_l V_k Y_{lk} cos(\delta_l - \delta_k + \theta_{lk}) \tag{15}$$

$$Q_l^{inj} = \sum_{j=1}^{N_{bs}} V_l V_k Y_{lk} sin(\delta_l - \delta_k + \theta_{lk}) \tag{16}$$

In the above equations, the injected real and reactive powers into buses are respectively shown by P_l^{inj} and Q_l^{inj}; in addition to that, N_{bs} denotes total buses of system; the voltage of buses l and k are respectively demonstrated by V_l and V_k; Y_{lk} and θ_{lk} signify the magnitude and phase of elements of admittance matrix; the angles of buses' voltage are respectively revealed by δ_l and δ_k.

- Soft open points:

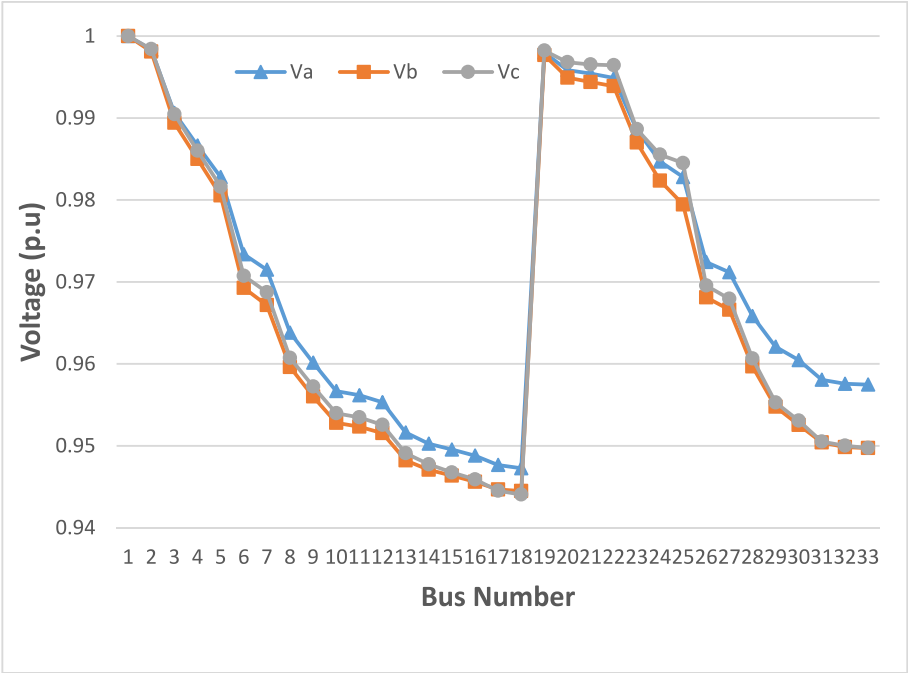


Fig. 16. The voltage profile of unbalanced network.

A SOP substitutes a tie-line of distribution networks, resulting in linking two buses [17,25]. This in turn gives this opportunity to exchange power among connected buses. To model SOPs, some constraints are must be satisfied. Constraint (17) satisfies the active power exchange between two adjacent feeders; the losses of each voltage source converters are also calculated by constraints (18) and (19); the reactive powers of SOP is kept between acceptable ranges using (20) and (21); the capacity limitation of the sop is also checked by (22) and (23).

$$P^{i}_{SOP} + P^{j}_{SOP} + P^{i}_{SOP,loss} + P^{j}_{SOP,loss} = 0 \tag{17}$$

$$P^{i,t}_{SOP,loss} = A^{i}_{SOP}\sqrt{\left(P^{i,t}_{SOP}\right)^2 + \left(Q^{i,t}_{SOP}\right)^2} \tag{18}$$

$$P^{j,t}_{SOP,loss} = A^{j}_{SOP}\sqrt{\left(P^{i,t}_{SOP}\right)^2 + \left(Q^{i,t}_{SOP}\right)^2} \tag{19}$$

$$Q^{i,min}_{SOP} \le Q^{i,t}_{SOP} \le Q^{i,max}_{SOP} \tag{20}$$

$$Q^{j,min}_{SOP} \le Q^{j,t}_{SOP} \le Q^{j,max}_{SOP} \tag{21}$$

$$\sqrt{\left(P^{i,t}_{SOP}\right)^2 + \left(Q^{i,t}_{SOP}\right)^2} \le S^{i,max}_{SOP} \tag{22}$$

$$\sqrt{\left(P^{i,t}_{SOP}\right)^2 + \left(Q^{j,t}_{SOP}\right)^2} \le S^{j,max}_{SOP} \tag{23}$$

where,P^{i}_{SOP}, P^{j}_{SOP} $P^{i}_{SOP,loss}$, and $P^{j}_{SOP,loss}$ are the active power transfers between two adjacent feeders and their corresponding losses, respectively.$A^{i,j}_{SOP}$ shows the proportion of power lost by SOP. $Q^{i,t}_{SOP}$ and $Q^{j,t}_{SOP}$ are the reactive powers which SOP is provided and are maintained between their permissible values by Q^{min}_{SOP} and Q^{max}_{SOP}. Also, $S^{i,max}_{SOP}$ and $S^{j,max}_{SOP}$ are the rating of voltage source converters.

3. Optimization algorithm

3.1. Particle swarm optimization

One of the most popular optimization algorithms is the PSO algorithm inspired by flying a group of birds [34–36]. Kennedy and Eberhart were the first to offer it in 1995. Because the method relies on a basic notion and it could be conveniently handled by computer systems, it has caught the interest of different scientists worldwide and has also been successfully cover a wide range of engineering challenges. Members in population are adjusted by recurring randomly to previously successful regions in the search space, and are impacted by the performances of other members of populations. To put this concept into words, each member of population refers to a possible solution. The position of each member is then updated based on its previous position, velocity, and other members in search space, with the aim of reaching a better position for itself. The process of PSO algorithm is described step by step as in the following:

Step 1: similar to other population-based optimization algorithms, there are some parameters in PSO which must be tuned before getting the problem solved. The parameters associated with PSO include total member of population/particles, maximum iteration for stopping the optimization (NI), cognitive and social coefficients (C1, C2), and inertia weight factor (w).

Step 2: after defining the necessary parameters in previous step, we are able to generate the stochastic solutions described as follows:

$$X_{i,j} = LB_j + \left(UB_j - LB_j\right)*rand[0,1]$$

$$i = 1,2,\ldots,PopMax\ ,\quad j = 1,2,\ldots,NV \tag{24}$$

Here, X is the member of population; LB and UB are the lower and upper boundaries of search space, respectively; *PopMax* and *NV* respectively signify the number of population and decision variables.

Step 3: this section has several subsections which are leveraged to make a new particle:

Step 3.1: the velocity of *ith* particle is updated based on the following concept:

$$V^{t+1}_{i,j} = wV^{t}_{i,j} + C_1\ rand[0,1] \times \left(Pbest^{t}_{i,j} - X^{t}_{i,j}\right) + C_2\ rand[0,1] \times \left(Gbest - X^{t}_{i,j}\right) \tag{25}$$

V is the velocity of the particles, the personal best and global best are shown by *Pbest* and *Gbest, respectively*.

Step 3.2: the new solution is created based on the updated velocity

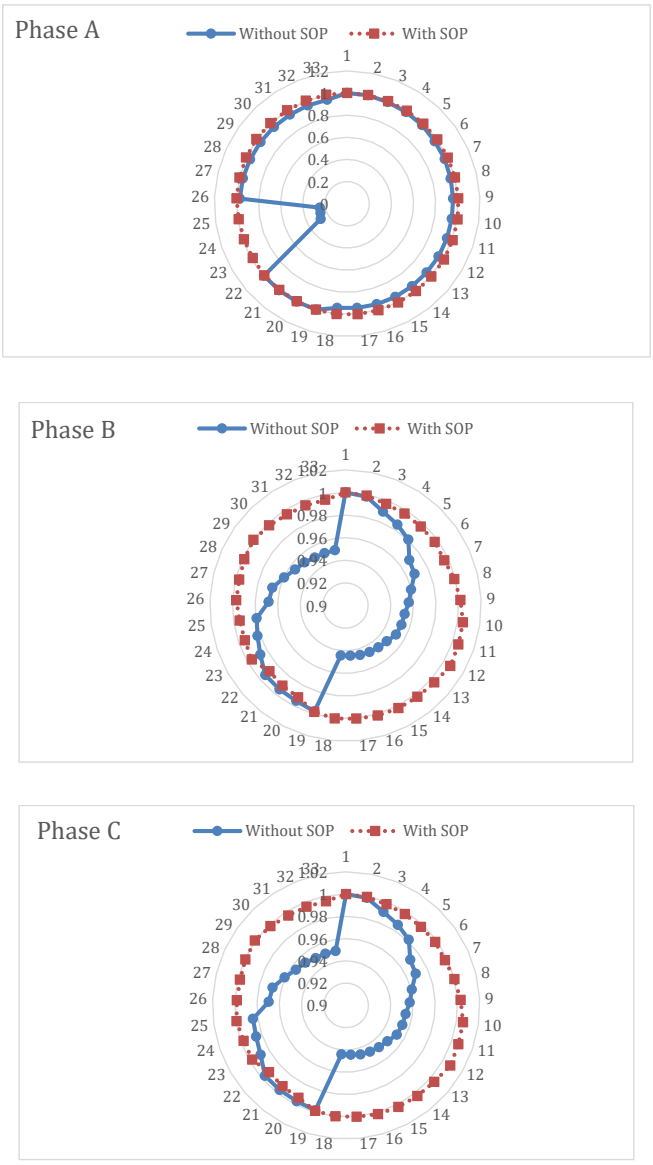


Fig.17. The voltage profile of unbalanced network for phases A-C, under fault conditions.

and position of the ith particle in population in previous iteration. This is described as follows:

$$X_{i,j}^{t+1} = V_{i,j}^{t+1} + X_{i,j}^{t} \tag{26}$$

Step 4: the particle (solution) generated in previous step is now evaluated on the basis of the objective function. Following that, the obtained fitness is compared to the fitness of the personal best (*Pbest*) and global best (*Gbest*) and thus the desired particles are kept for the next iteration as given:

if $fit(X_{i,\mathrm{NV}}^{t+1}) < fit(Pbest_{i,\mathrm{NV}}^{t})$ then,

$$Pbest_{i,\mathrm{NV}}^{t} = X_{i,\mathrm{NV}}^{t+1} \tag{27}$$

end

if $fit(Pbest_{i,\mathrm{NV}}^{t}) < fit(Gbest)$ then,

$$Gbest = Pbest_{i,\mathrm{NV}}^{t} \tag{28}$$

end

Step 5: here, maximum number of iterations is the stopping criteria, meaning that steps 3 and 4 are repeated until reaching maximum number of iterations.

3.2. The steps of PSO to solve the proposed model

Here, the particles are candidate to optimally schedule the SOP in distribution networks. To this aim, population are firstly generated between the upper and lower bounds of the problem, which are the available power provided by SOPs. After that, particles are updated based on the velocities and their previous experiences and their fitness are measured using objective function (here, VSF). Next, the personal and global bests are updated by the best fitness. This process is repeated until meeting the maximum iteration. To easily understand this process, the flowchart of the proposed approach are shown in Fig. 2.

4. Simulation set up and results

The test network of this investigation is depicted in Fig. 3a which is utilized to check the suggested methodology's efficacy. The problem formulas are coded in MATLAB environment. The total active and reactive demands of this network are respectively 3.7 MW and 2.3 MVAR with nominal voltage of 12.66 kV [37,38]. There are two SOPs in this network which connect bus 12 to 22 and 25 to 29, respectively based on the reference [39]. The structure of SOP is shown in Fig. 3b. Two VSCs are connected back-to-back via a DC link capacitor. Each VSC connect to the specified bus via a series filter (inductance L). This inductance filters the high-frequency harmonics and facilitates control of power flow. The sizing of SOPs could be achieved in planning step according to the network situations and operator preferences. For example, some operators just considered the investment of SOPs, while other may concentrate on the technical aspects. In this investigation, we found 1MVA capacity for SOPs based on the objective function (voltage stability enhancement) and then they are scheduled based on the condition of the network to reach the better voltage stability. Also, 4 photovoltaic resources which their generation profiles over a day have been shown in Fig. 4 are installed at buses 7, 10 24, and 27, respectively. Two wind turbines are also located on the buses 13 and 30 and their generations are illustrated in Fig. 5. The proposed method is checked for 24 h which the load factor is depicted in Fig. 6, which means, these load factors multiple to the load demand of network. The developed framework is also assessed under 4 case studies listed as flows:

Case1- The base case where there are no renewable energy sources and SOPs.

Case2- Considers just renewable energy resources in the base case.

Case3- Considers only SOPs in the base case.

Case4- Considers both SOPs and renewable energies into the base case.

After obtaining the results of different case studies, comparisons are provided here. To begin with, the minimum VSF of network at each hour for 4 cases are depicted in Fig. 7. Due to the fact that higher VSF means more stable network, the higher VSF is more desirable. As can be seen from this figure, VSF rate in case 1 varies approximately between 0.9 and 0.94 per unit which was the minimum quantity over the entire time frame. Apart from hours 19–24, which this trend roughly remained steady, VSF quantity was slightly increased to almost 0.9 and 0.95 per unit in case 2, where renewable energy resources are merely integrated into network. Interestingly, SOPs have had a significant impact on improving VSF of network, so that case 3 can be ranked as the second-best method. In sharp contrast to cases 1–3, the combination of renewable resources and SOPs provided the higher VSF, where the peak is reached at around 0.98 per unit. From this comparison, it can be observed that the network stability experiences substantial enhancement by means of managing the SOPs.

A question may be arisen how SOP improves the voltage of the network. Conventional distribution networks are generally radial and one-directional power flow, meaning that they are supplied by the substation in bus 1. As a result of this, there is voltage drop in buses

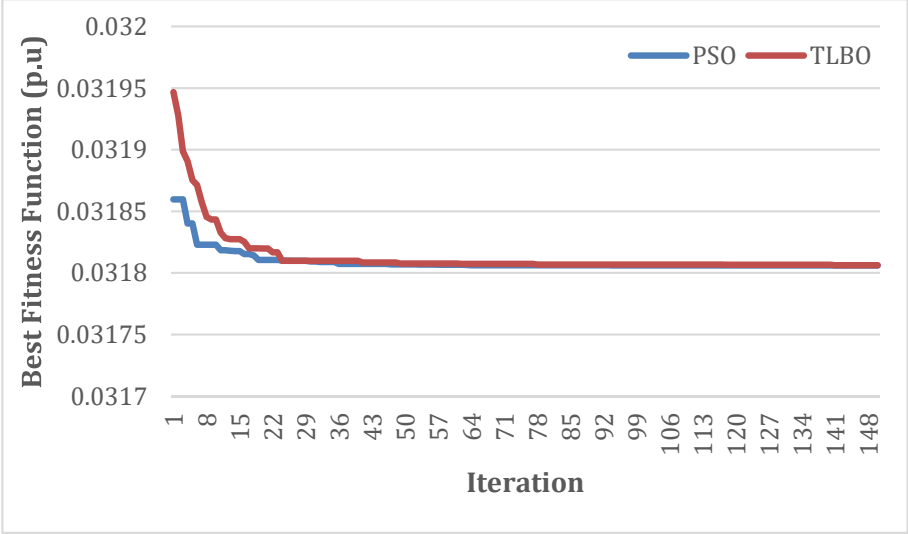


Fig.18. The performance characteristics of PSO versus TLBO.

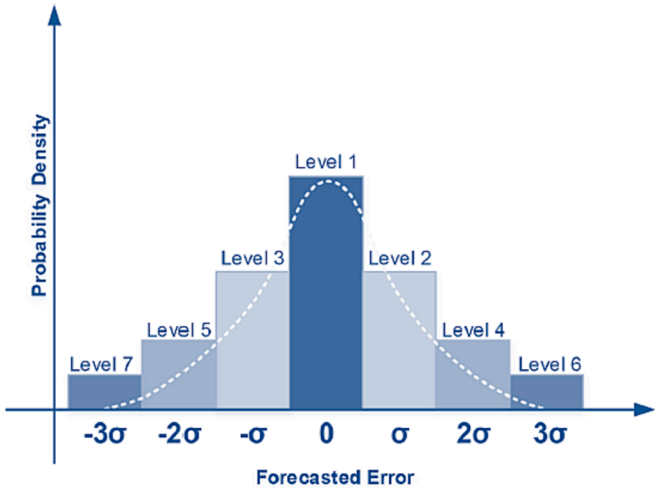


Fig. 19. The normal PDF with 7 segments.

which are far away from the bus 1. However, if two adjacent feeders are connected together by SOP, the active power passing through the adjacent feeders can be controlled which leads to have a controllable power flow among feeders to tackle the network problems like voltage deviation and power loss minimization. Besides, SOPs are designed by power converters with the ability of reactive power support as well. As a result of this, the reactive power of feeders is also compensated by the SOP management which has a direct effect on voltage management of distribution networks. Here, in this investigation, an optimization approach is developed to effectively manage the powers provided by SOPs to improve the voltage stability under normal operation and network fault conditions. In other words, the proposed optimization approach searches to find the most appropriate reference signals used in the control system of SOPs to improve the voltage stability. It is important to note that in normal operation, the DC link voltage and the reactive power of one side of the SOP are controlled by one of the VSCs, and the other VSC is responsible for controlling the active power flowing through two adjacent feeders and the reactive power of the other side of the SOP. However, in network fault conditions and disconnection of one of the VSCs from the grid, the corresponding VSC is controlled in grid forming mode to feed the isolated loads and the other VSC controls the DC link voltage and the reactive power of other side of the SOP. Interested readers can refer to [40] to study the details of the control system of a SOP.

Despite comparing the amount of VSF provided in prior paragraph, the impact of SOP on voltage stability is discussed here by using the P-V and P-Q curves. In order to calculate the mentioned curves, the load at each bus of network is increased and the voltage of buses are measured [41]. If each case study could tolerate higher value of load, it means it has better voltage stability margin. Figs. 8 and 9 illustrate these curves for bus 30 of network. From these figures, it is obvious that getting SOPs as a method for voltage stability improvement has also ranked as the best technique due to its active role in increasing voltage stability margin, leading to accommodating more demand and defer the updating of aged infrastructure of distribution networks.

Let's have a look at the power loss of the network. The summation of real power loss on the daily basis is depicted on Fig. 10 for cases 1–4. Here, we are looking to recognize what case study minimizes the most power loss, which means the more energy saving. As can be observed, the amount of power loss is maximum in the case 1, where no resources are included. Nevertheless, there is a slight decrease in the power loss in case 2 compared to case 1 because of integration of renewable energies. Interestingly, in case 3, just through SOPs management, the power loss is decreased doubly. Finally, power loss is plunged to its minimum value in case 4 due to simultaneous integration of renewable resources and SOPs. From this result, it can be concluded that if SOPs integrate in the network, a significant amount of power loss decreases. This highlights the positive influential of SOPs in distribution grids.

Figs. 11 and 12 show the transferred active powers by SOPs and the amount of reactive power provided by SOPs in case 3 and 4, respectively. In terms of transferred active powers between feeders, it is clear that while active power is injected to one bus, it is absorbed from another bus. For example, SOP, connecting bus 12 to 22, injects power into bus 22 by absorbing power from bus 22. However, this is vice versa for SOP linked buses 25 to 29. It is notable that the idle capacity of SOPs has also been used to support reactive power of network, too. It is worthwhile to be noted that, SOP can interestingly help to moderately control the power between feeders without changing the configuration of the network.

The previous experiments were implemented on the test network under normal operation condition. However, the potential of SOPs in supporting the network under fault condition is evaluated here. Distribution networks are designed as ring, while they are operated radially on account of the cutting back the disturbances between protection devices [42]. Accordingly, there is one connection between the feeders

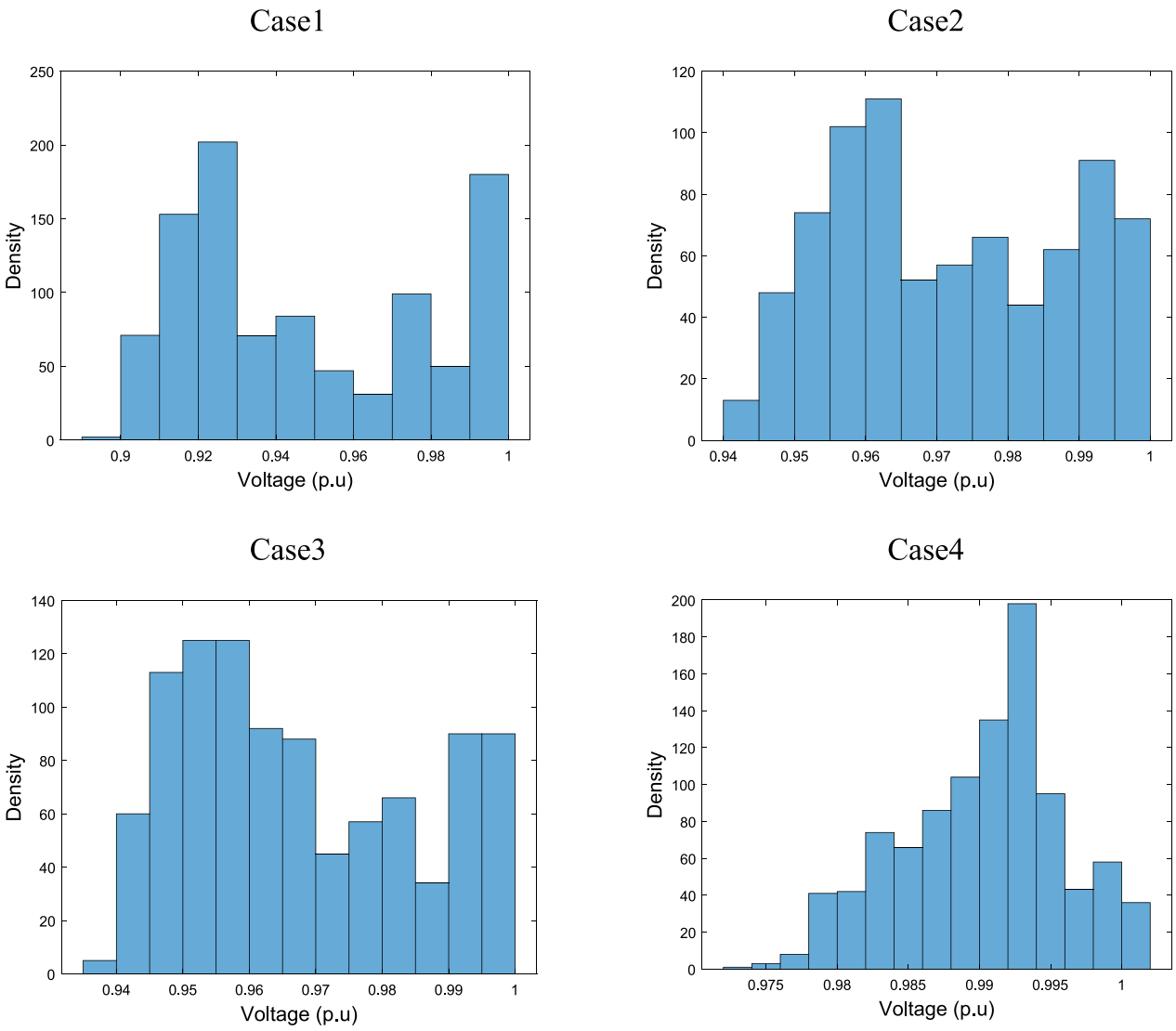


Fig. 20. The histogram of network's voltage for all case studies.

and grids and thus if a branch is disconnected from the network by natural disasters like hurricanes, floods, etc., that part is outage and cannot be supplied. After the occurrence of such incidents, the stability of network witnesses a sudden decline. To this end, the possibility of SOP to compensate for this phenomenon should be assessed. Consider the 33-bus network depicted in Fig. 13 as the under-test network. From this figure, it is assumed that line 22, connecting bus 3 to 23, is damaged and therefore buses 23–25 are disconnected from the network. To have a distinct comparison, the network is operated under two conditions, including without SOPs and with SOPs and the results are provided.

The voltage of the network under the fault for two cases, without SOPs and with SOPs is demonstrated in Fig. 14. As this figure implicates, the voltage of network for separated buses from the network (23–25) plunged to approximately zero when there are no SOPs integration in the network. On the contrary, upon effectively managing SOPs, this problem thoroughly solved and the voltages of mentioned buses brought back to the acceptable values, accounting for virtually 1p.u. Let's have a glance on the voltage stability factor of the network. The VSF of the network for without/with SOPs is illustrated in Fig. 15. The comparison reveals that the VSF collapsed dramatically, where the SOPs are not dedicated into the network. Nonetheless, there is a substantial improvement in the network's stability whether SOPs are managed appropriately. The comparison divulges that the SOP have appropriate effects of supporting the network, so the stability of network could be maintained among the permissible values to have a reliable network, resulting in delivering the stable power to the customers. Technically, when a fault occurs, some customers may be disconnected from the main grid and experience power outage. To solve this problem, by using SOP isolated customers can be supplied through another feeder connected to SOP. The advantage of this strategy can be highlighted as a resilience enhancement of distribution networks through SOPs.

Despite the balanced network under fault situation, the unbalanced network is also taken into account to point the potential of SOPs out. The data associated with this unbalanced network was extracted from reference [43]. In order to have a clear overview on the unbalanced voltage profile, the voltage profile under unbalanced network have been depicted in Fig. 16 for 0.6p.u load level. It is clear from the voltage profile that the considered network has been unbalanced due to load imbalance. The voltage profile of each phase has been plotted in Fig. 17 when a fault cause disconnecting of the phase A in line 22, connecting bus 3 to 23. From the result, it is obvious that SOPs could be utilized to enhance the resilience of the power distribution networks under both balanced and unbalanced circumstances.

The evaluation of PSO's performance is also another factor which should be deemed by comparing with other algorithms. In this regard, the PSO is compared with teaching–learning-based optimization (TLBO)

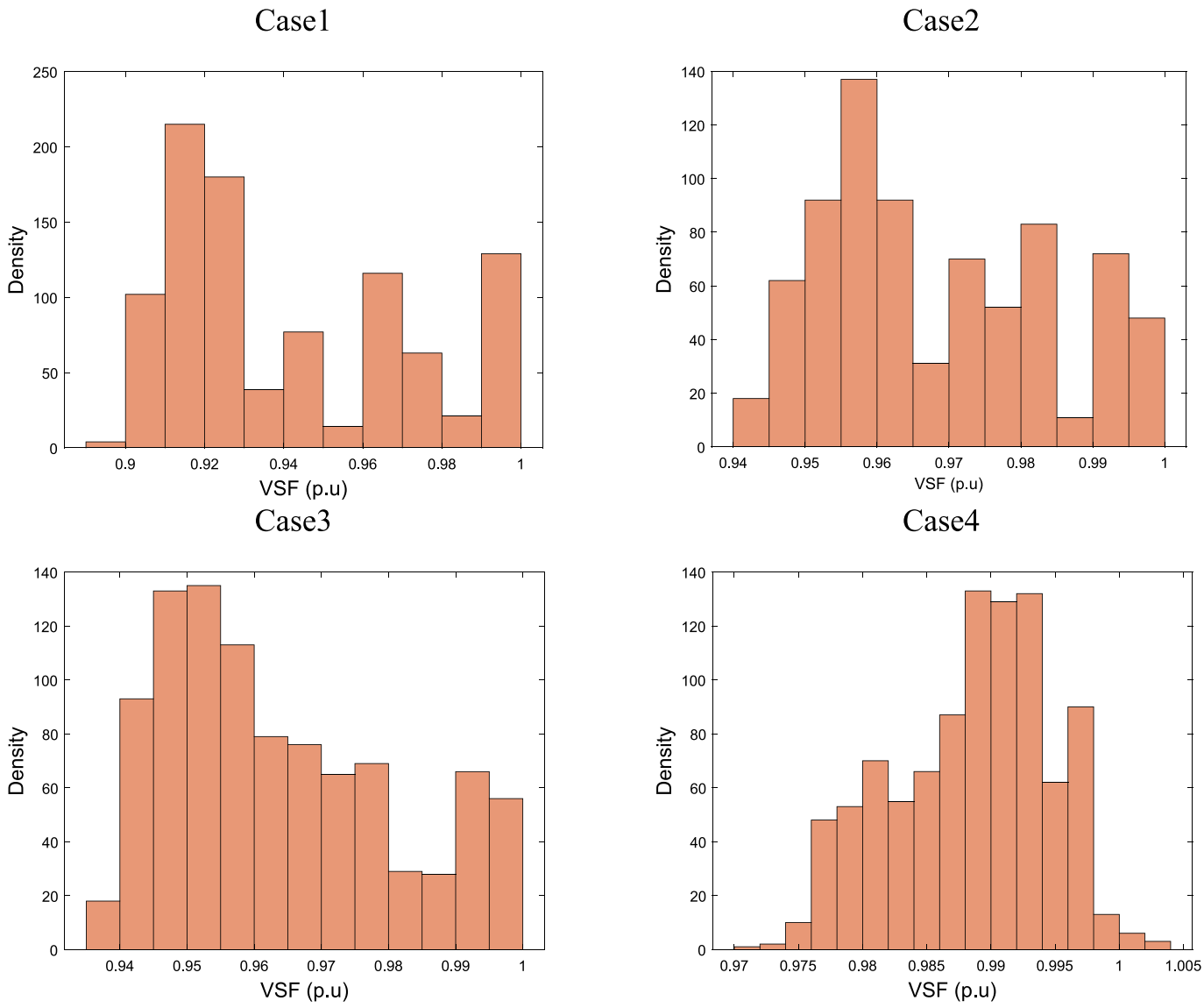


Fig. 21. The histogram of network's vsf for all case studies.

Table 1
Comparison of the proposed method with previous work.

Scenarios	Power loss (kW)	Min voltage (p. u)	Min VSF (p. u)
Scenario1: Base case (without SOP)	202.8	0.913	0.911
Scenario2: (extracted from ref [20])	124.4	0.934	0.932
Scenario3: the proposed method	155.6	0.948	0.947

[44] and the convergences of both algorithms are depicted in Fig. 18. From this figure, it can be seen that PSO algorithm found the satisfying result in approximately 19 iterations. In sharp contrast to this, this trend accounted for almost 25 iterations for TLBO. From this, it is obvious that PSO and TLBO are able to find the appropriate solution, but PSO may be superior over TLBO a little because of its better search capability which led to converging in 19 iterations while TLBO needs 6 iterations more. Overall, both algorithms have the great performance and this comparison confirms the efficiency of proposed optimization tools to manage SOPs.

Due to the fact that power systems are under the uncertainties related to the renewable resources and loads, its effect should be investigated, resulting in making the model much more realistic. Towards this end, scenario-based technique which has been completely discussed in [45,46] is utilized to model the uncertainties as discussed as follows:

Initially, a suitable probability density function (PDF) is selected to convey the unknown parameters. The PDFs are then split up into several sections. Fig. 19, for instance, shows a typical PDF with seven components. Specific errors in the unknown parameter are represented by these probability levels in the PDF. A roulette wheel mechanism is used to generate scenarios. A random integer between 0 and 1 is produced for each ambiguous variable. This arbitrary integer has a probability level assigned to it in the PDF. Up till a collection of situations is formed, this procedure is repeated. Following the generation of the initial scenarios, scenarios with low probability or those that are identical to others are ignored. This is done to lessen the computational strain. The generated set of scenarios is used as input data for the optimization problem.

In this investigation, a set of scenarios, here 1000 scenarios, are generated and then reduced to 30 scenarios for decreasing the computational burden. After solving the problem for reduced scenarios, the results could be provided as histogram to have better comparison. In

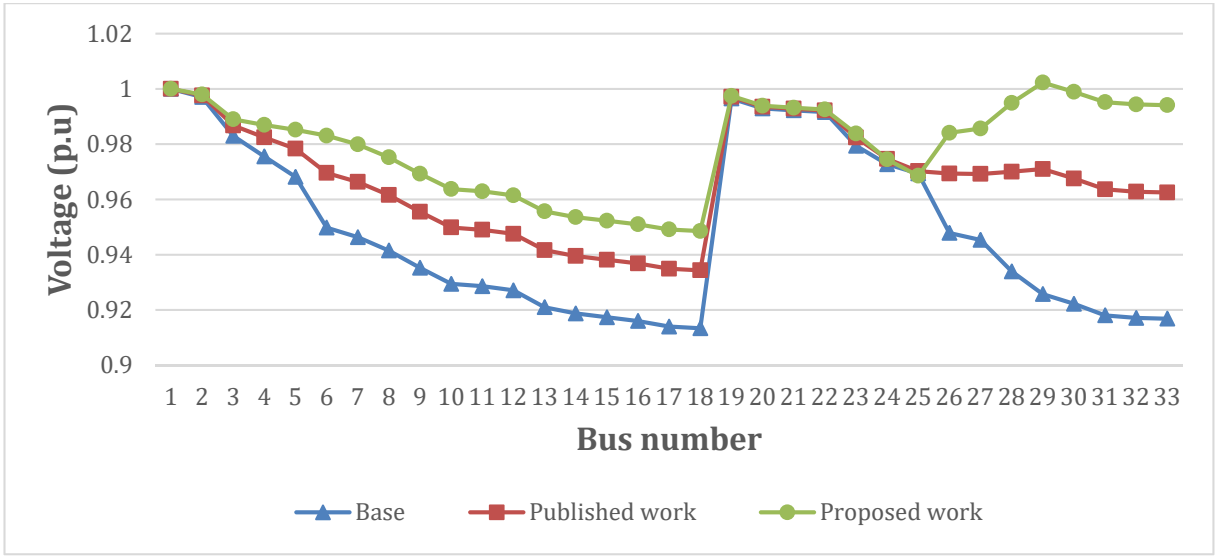


Fig. 22. The voltage profile of the proposed method vs existing work in the literature.

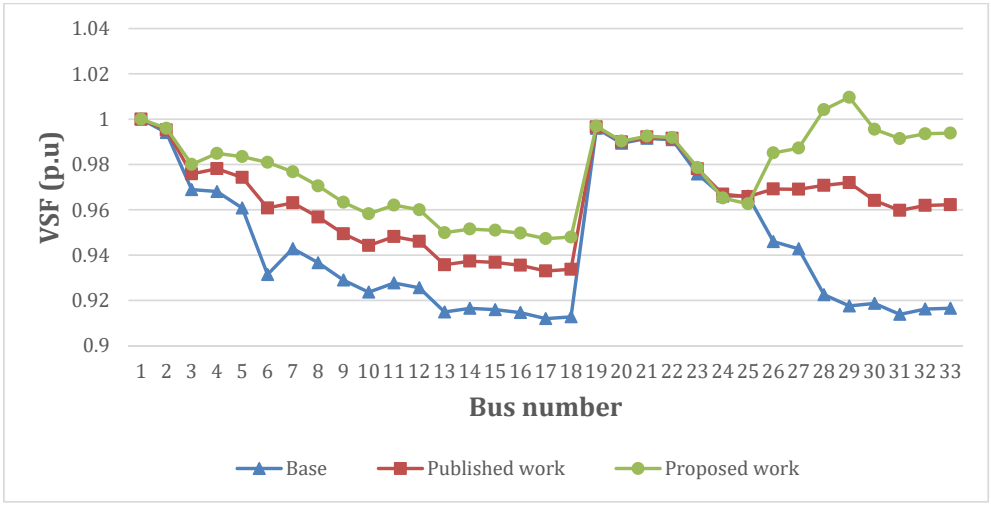


Fig. 23. The vsf of the proposed method vs existing work in the literature.

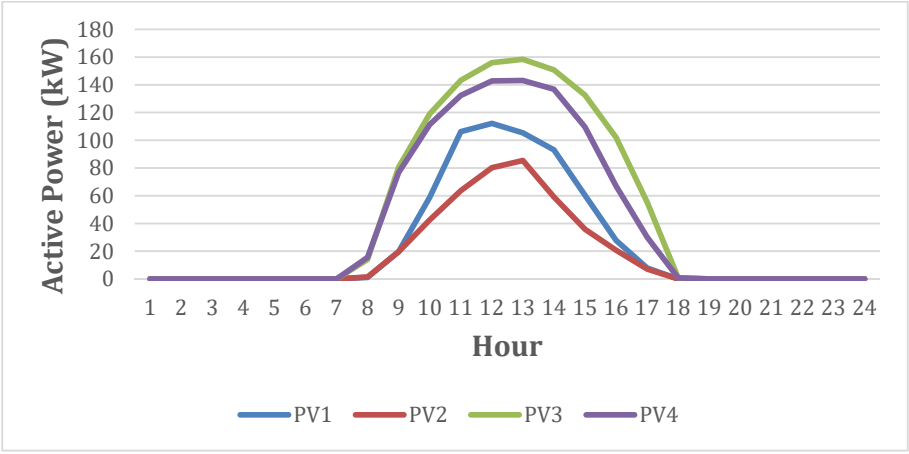


Fig. 24. The forecasted values of pvs in 69-bus network.

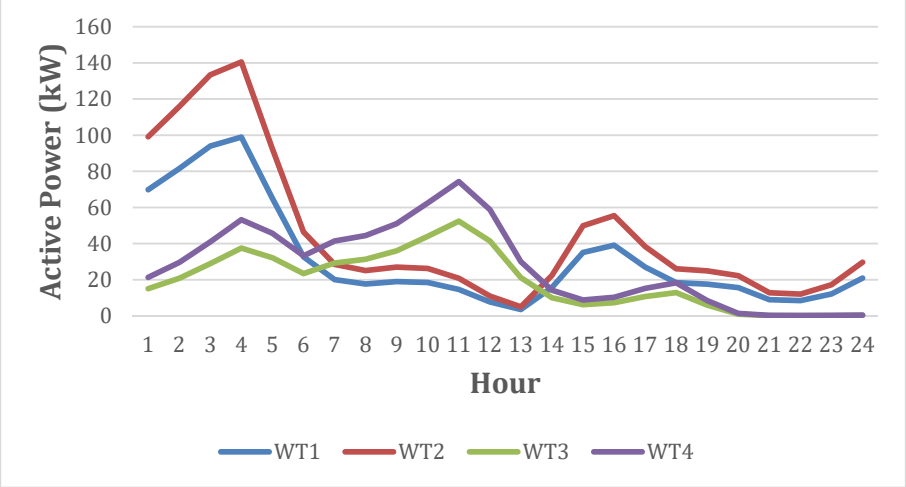


Fig. 25. The forecasted values of wts in 69-bus network.

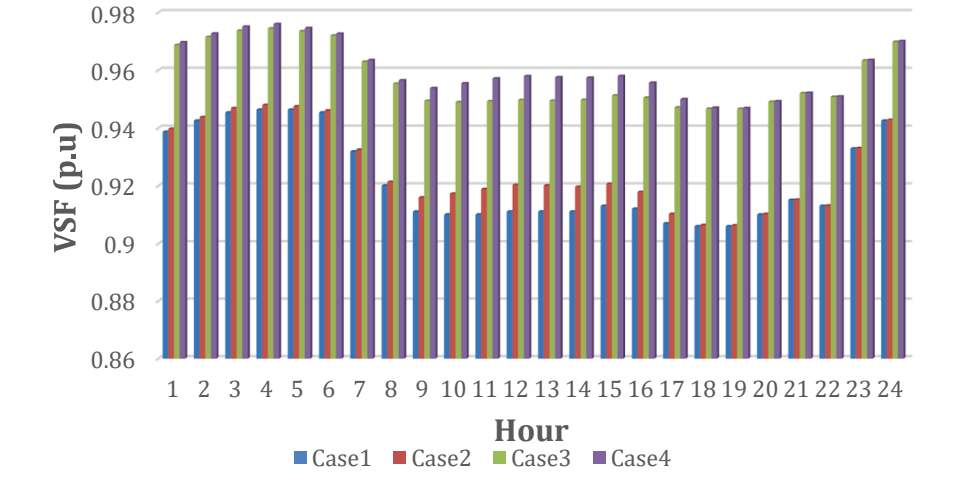


Fig. 26. The minimum vsf for 69-bus network.

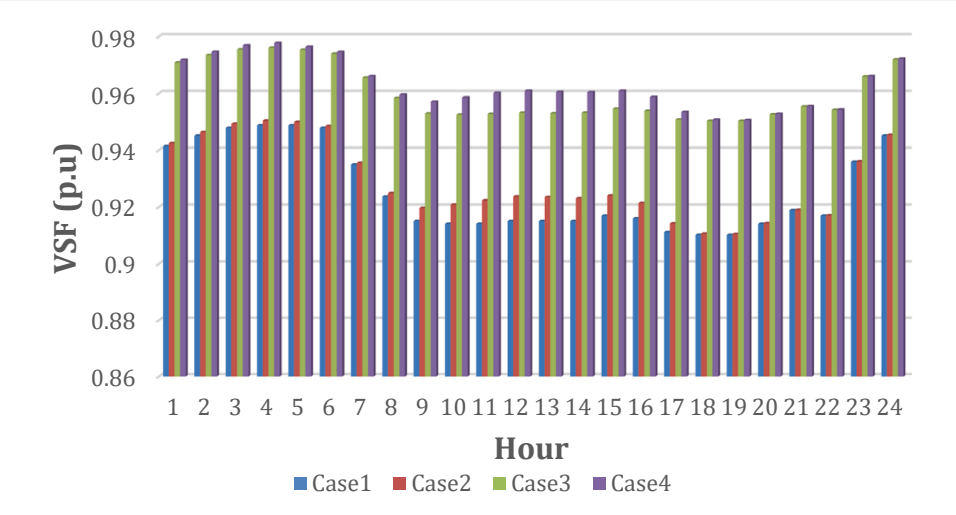


Fig. 27. The minimum voltage of the 69-bus network.

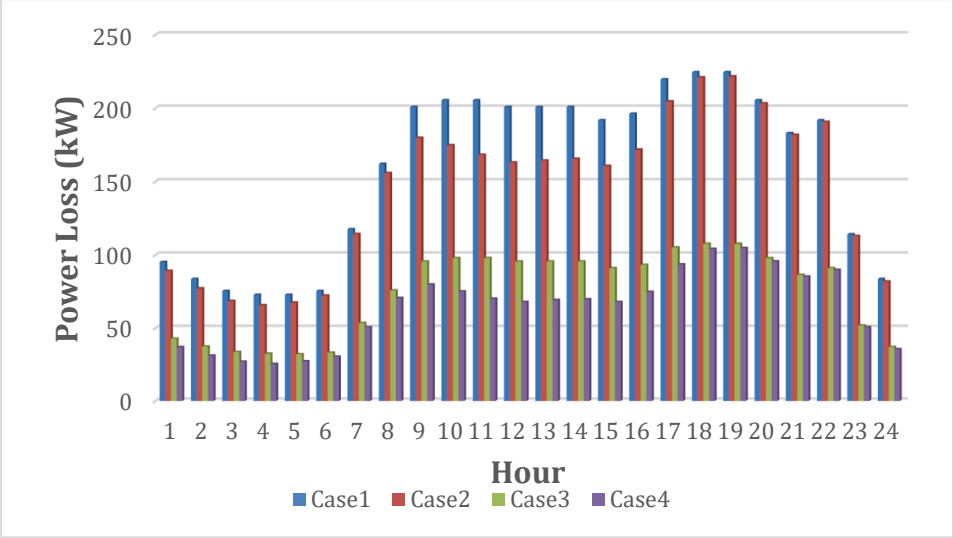


Fig. 28. The power loss of 69-bus network.

terms of results, the histogram of the network's voltage for cases are shown in Fig. 20. As clearly seen, voltage of the grid is kept within the permissible ranges in case4, where RESs and SOPs are integrated into the network. As a result, the SOP management could be a great remedy to tackle the violation of uncertain parameters and have flatter voltage over all buses of the network.

Similarly, the amount of VSF for all cases is depicted on the Fig. 21. From this, it can be observed that case 4 witnessed a better VSF compared to other cases, meaning that more stable voltage. From this explanation, it is clear that SOP is a promising strategy to enhance the voltage stability of power distribution grids under high penetration of renewable resources.

Another way to challenge the proposed method is to compare the proposed method with existing work in the literature. Towards this attained, the method used in reference [20] is compared with the proposed method. It is worth mentioning that all test conditions are the same for both methods and the only difference is in their objective function. In other words, in the mentioned (comparable) method, the objective function is to reduce network losses, but the objective function of this investigation is to improve the network stability index. The results of comparing the proposed method with previous works are shown in Table 1. Based on this table, three different scenarios have been taken into account, which include: the network without soft open points, the reference results [20] and the results obtained with the proposed method in this research. Based on the results in the mentioned table, the losses for the first scenario to the third scenario are equal to 202.8, 124.4 and 155.4 kW, respectively. From the comparison of network losses, it can be concluded that the best case is related to the second scenario. Although the proposed method (scenario3) has more losses than the second scenario, but in the proposed method, a significant amount of network losses has been reduced compared to the base case. The next parameter that we must examine is the minimum network voltage, which for the first scenario to the third scenario are equal to: 0.913, 0.934 and 0.948p.u, respectively. It is clear from this comparison that the proposed method has provided better results among other scenarios. The next parameter to be compared is the minimum network stability index, which for the first scenario to the third scenario is equal to: 0.911, 0.932 and 0.947p.u, respectively. Once again, the proposed method has provided better results to improve the voltage stability of the network.

In order to be able to compare the network voltage for all different buses in the first to third scenarios, the voltage profile and stability index for all buses are shown in Fig. 22 and Fig. 23, respectively. As it is evident, the proposed method has provided acceptable and better results than previous work.

Despite the 33-bus network, the proposed strategy is also implemented on the 69-bus network [38]. The 69-bus network includes 4 PVs which were installed on buses 20, 32, 61 and 69, respectively based on the forecasted values are shown in Fig. 24. Similarly, the forecasted values of wind turbines which are allocated on buses 18, 25, 42, and 53 are illustrates in Fig. 25. Two SOPs have also been integrated into network which connect bus 15 to 46 and 27 to 65, respectively. This network is similarly assessed under four case studies like 33-bus network and the results are provided. In terms of VSF, Fig. 26 has been depicted. As can be observed, the proposed method has a tremendous effect on improving the voltage stability of the network, so that the network experienced the best situation in case 4, where both SOPs and renewables are integrated simultaneously. Fig. 27 divulges the amount of minimum voltage of the network per hour. As can be seen, SOPs can be a great remedy to deal with voltage issues in power distribution networks, particularly in case 4 which renewables are also considered. The power loss of the network has been demonstrated in Fig. 28. From this figure, it is obvious that the network experienced a significant power loss reduction whether SOPs and renewables are integrated to the network. From this explanation, it can be deduced that SOP is a viable remedy to support current distribution networks to share the power among feeders which leads to hosting more renewable integration to cope with climate change.

5. Conclusion

Nowadays, thanks to the development in power electronic-based equipment, secure operation of power systems has experienced a great betterment. This paper, for the first time, analyzed the potential of soft open points, to improve the voltage stability of distribution networks integrated with solar and wind energies. As can be observed from different case studies, soft open points take significant part in improving the voltage stability of network which leads to higher stability margin. The power loss has also been evaluated under different cases. It was obvious that it had also a substantial impact on power loss minimization, approximately 50 percent compared with base case. Voltage stability assessment under the fault condition has also been analyzed. It was vivid that by using SOPs, the voltage and stability factor of network are improved substantially and are kept between the standard situations. From this investigation, soft open points could enable operators to manage both active and reactive powers between feeders which improve the efficiency of the network. There is no doubt that each method has

limitations, despite its positive effect on the network. In this method, it should be noted that the formulation of this paper was developed in a nonlinear form, which may make it a little time-consuming compared to the linear programming. Accordingly, in the future work, we will do our best to develop a linear formulation to reach out the commercial solvers like CPLEX and GUROBI. Due to the fact that SOP has recently been introduced, there are still various topics for future works, such as voltage unbalance compensation, voltage sag improvement, etc.

CRediT authorship contribution statement

Kambiz Mardanimajd: Investigation, Software. **Shahram Karimi:** Conceptualization, Data curation, Supervision. **Amjad Anvari-Moghaddam:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Berrang-Ford L, Ford JD, Paterson J. Are we adapting to climate change? Glob. Environ. Chang. 2011;21(1):25–33. https://doi.org/10.1016/j.gloenvcha.2010.09.012.

[2] Beard LM, et al. Key technical challenges for the electric power industry and climate change. IEEE Trans. Energy Convers. 2010;25(2):465–73. https://doi.org/10.1109/TEC.2009.2032578.

[3] Olabi AG, Abdelkareem MA. Renewable energy and climate change. Renew. Sustain. Energy Rev. Apr. 2022;158:112111. https://doi.org/10.1016/j.rser.2022.112111.

[4] Chandramowli SN, Felder FA. Impact of climate change on electricity systems and markets - a review of models and forecasts. Sustain. Energy Technol. Assessments 2014;5:62–74. https://doi.org/10.1016/j.seta.2013.11.003.

[5] Fawzy S, Osman AI, Doran J, Rooney DW. Strategies for mitigation of climate change: a review. Environ. Chem. Lett. 2020;18(6):2069–94. https://doi.org/10.1007/s10311-020-01059-w.

[6] Gholami K, Azizivahed A, Arefi A. Risk-oriented energy management strategy for electric vehicle fleets in hybrid AC-DC microgrids. J. Energy Storage Jun. 2022;50:104258. https://doi.org/10.1016/j.est.2022.104258.

[7] Craig MT, et al. A review of the potential impacts of climate change on bulk power system planning and operations in the United States. Renew. Sustain. Energy Rev. 2018;98:255–67. https://doi.org/10.1016/j.rser.2018.09.022.

[8] Ludig S, Haller M, Schmid E, Bauer N. Fluctuating renewables in a long-term climate change mitigation strategy. Energy 2011;36(11):6674–85. https://doi.org/10.1016/j.energy.2011.08.021.

[9] Luo X, Liu Y, Liu J, Liu X. Optimal design and cost allocation of a distributed energy resource (DER) system with district energy networks: a case study of an isolated island in the South China Sea. Sustain. Cities Soc. 2019;51:101726. https://doi.org/10.1016/j.scs.2019.101726.

[10] Saadati R, Jafari-Nokandi M, Saebi J. Allocation of RESs and PEV fast-charging station on coupled transportation and distribution networks. Sustain. Cities Soc. 2021;65. https://doi.org/10.1016/j.scs.2020.102527.

[11] HassanzadehFard H, Jalilian A. Optimal sizing and location of renewable energy based DG units in distribution systems considering load growth. Int. J. Electr. Power Energy Syst. Oct. 2018;101:356–70. https://doi.org/10.1016/j.ijepes.2018.03.038.

[12] Parizad A, Hatziadoniu K. Security/stability-based Pareto optimal solution for distribution networks planning implementing NSGAII/FDMT. Energy Feb. 2020;192:116644. https://doi.org/10.1016/j.energy.2019.116644.

[13] Parihar SS, Malik N. Optimal allocation of renewable DGs in a radial distribution system based on new voltage stability index. Int. Trans. Electr. Energy Syst. Apr. 2020;30(4):e12295.

[14] Yang CF, Lai GG, Lee CH, Su CT, Chang GW. Optimal setting of reactive compensation devices with an improved voltage stability index for voltage stability enhancement. Int. J. Electr. Power Energy Syst. May 2012;37(1):50–7. https://doi.org/10.1016/j.ijepes.2011.12.003.

[15] Parihar SS, Malik N. Optimal integration of multi-type DG in RDS based on novel voltage stability index with future load growth. Evol. Syst. Dec. 2021;12(4):981–95. https://doi.org/10.1007/s12530-020-09356-z.

[16] B. Ismail, M. M. Naain, N. I. A. Wahab, L. J. Awalin, I. Alhamrouni, and M. F. A. Rahim, "Optimal placement of DSTATCOM in distribution network based on load flow and voltage stability indices studies," in *2017 International Conference on Engineering Technology and Technopreneurship, ICE2T 2017*, 2017, vol. 2017-Janua, pp. 1–6, doi: 10.1109/ICE2T.2017.8215973.

[17] Fuad KS, Hafezi H, Kauhaniemi K, Laaksonen H. Soft open point in distribution networks. IEEE Access 2020;8:210550–65. https://doi.org/10.1109/ACCESS.2020.3039552.

[18] Tavakoli Ghazi Jahani MA, Nazarian P, Safari A, Haghifam MR. Multi-objective optimization model for optimal reconfiguration of distribution networks with demand response services. Sustain. Cities Soc. 2019;47:101514. https://doi.org/10.1016/j.scs.2019.101514.

[19] Pamshetti VB, Singh SP. Coordinated allocation of BESS and SOP in high PV penetrated distribution network incorporating DR and CVR schemes. IEEE Syst. J. 2020. https://doi.org/10.1109/JSYST.2020.3041013.

[20] Cao W, Wu J, Jenkins N, Wang C, Green T. Benefits analysis of Soft Open Points for electrical distribution network operation. Appl. Energy Mar. 2016;165:36–47. https://doi.org/10.1016/j.apenergy.2015.12.022.

[21] Long C, Wu J, Thomas L, Jenkins N. Optimal operation of soft open points in medium voltage electrical distribution networks with distributed generation. Appl. Energy 2016;184:427–37. https://doi.org/10.1016/j.apenergy.2016.10.031.

[22] Qi Q, Wu J. Increasing distributed generation penetration using network reconfiguration and soft open points. Energy Procedia 2017;105:2169–74. https://doi.org/10.1016/j.egypro.2017.03.612.

[23] W. Cao, J. Wu, and N. Jenkins, "Feeder load balancing in MV distribution networks using soft normally-open points," *IEEE PES Innov. Smart Grid Technol. Conf. Eur.*, vol. 2015-Janua, no. January, pp. 1–6, 2015, doi: 10.1109/ISGTEurope.2014.7028874.

[24] Sarantakos I, Zografou-Barredo NM, Huo D, Greenwood D. A reliability-based method to quantify the capacity value of soft open points in distribution networks. IEEE Trans. Power Syst. 2021;36(6):5032–43. https://doi.org/10.1109/TPWRS.2021.3071088.

[25] Pamshetti VB, Singh S, Thakur AK, Singh SP. Multistage coordination Volt/VAR control with CVR in active distribution network in presence of inverter-based DG units and soft open points. IEEE Trans. Ind. Appl. 2021;57(3):2035–47. https://doi.org/10.1109/TIA.2021.3063667.

[26] Cao W, He J, Zhou Y. Voltage control in distribution networks with soft open point and on-load tap changer. 2021 IEEE 4th Int. Conf. Electron. Technol. ICET 2021;2021:446–51. https://doi.org/10.1109/ICET51757.2021.9451001.

[27] Zhao J, Wang Y, Song G, Li P, Wang C, Wu J. Congestion management method of low-voltage active distribution networks based on distribution locational marginal price. IEEE Access 2019;7:32240–55. https://doi.org/10.1109/ACCESS.2019.2903210.

[28] Saboori H, Heshmati Y. Novel load partitioning and prioritization for distribution restoration with storage integrated-soft open point and energy-constrained resources. Electr. Power Syst. Res. 2023;214. https://doi.org/10.1016/j.epsr.2022.108947.

[29] Wang Y, Su X, Song M, Jiang W, Shahidehpour M, Xu Q. Sequential load restoration with soft open points and time-dependent cold load pickup for resilient distribution systems. IEEE Trans. Smart Grid 2023. https://doi.org/10.1109/TSG.2023.3262199.

[30] Kayal P, Chanda CK. Placement of wind and solar based DGs in distribution system for power loss minimization and voltage stability improvement. Int. J. Electr. Power Energy Syst. 2013;53:795–809. https://doi.org/10.1016/j.ijepes.2013.05.047.

[31] Moradi MH, Abedini M. A novel method for optimal DG units capacity and location in Microgrids. Int. J. Electr. Power Energy Syst. 2016;75:236–44. https://doi.org/10.1016/j.ijepes.2015.09.013.

[32] Hemmati M, Mohammadi-Ivatloo B, Ghasemzadeh S, Reihani E. Risk-based optimal scheduling of reconfigurable smart renewable energy based microgrids. Int. J. Electr. Power Energy Syst. 2018;101(October):415–28. https://doi.org/10.1016/j.ijepes.2018.04.005.

[33] Mohamed MA, Abdullah HM, El-Meligy MA, Sharaf M, Soliman AT, Hajjiah A. A novel fuzzy cloud stochastic framework for energy management of renewable microgrids based on maximum deployment of electric vehicles. Int. J. Electr. Power Energy Syst. Jul. 2021;129:106845. https://doi.org/10.1016/j.ijepes.2021.106845.

[34] Suman GK, Guerrero JM, Roy OP. Optimisation of solar/wind/bio-generator/diesel/battery based microgrids for rural areas: A PSO-GWO approach. Sustain. Cities Soc. Apr. 2021;67:102723. https://doi.org/10.1016/j.scs.2021.102723.

[35] Alshammari N, Asumadu J. Optimum unit sizing of hybrid renewable energy system utilizing harmony search, Jaya and particle swarm optimization algorithms. Sustain. Cities Soc. Sep. 2020;60:102255. https://doi.org/10.1016/j.scs.2020.102255.

[36] Gholami K, Dehnavi E. A modified particle swarm optimization algorithm for scheduling renewable generation in a micro-grid under load uncertainty. Appl. Soft Comput. May 2019;78:496–514. https://doi.org/10.1016/J.ASOC.2019.02.042.

[37] Bagheri Tolabi H, Ali MH, Rizwan M. Simultaneous reconfiguration, optimal placement of DSTATCOM, and photovoltaic array in a distribution system based on fuzzy-aco approach. IEEE Trans. Sustain. Energy 2015;6(1):210–8. https://doi.org/10.1109/TSTE.2014.2364230.

[38] Gholami K, Parvaneh MH. A mutated salp swarm algorithm for optimum allocation of active and reactive power sources in radial distribution systems. Appl. Soft Comput. 2019:105833. https://doi.org/10.1016/J.ASOC.2019.105833.

[39] Li P, et al. Coordinated Control method of voltage and reactive power for active distribution networks based on soft open point. IEEE Trans. Sustain. Energy Oct. 2017;8(4):1430–42. https://doi.org/10.1109/TSTE.2017.2686009.

[40] Cao W, et al. Operating principle of Soft Open Points for electrical distribution network operation. Appl. Energy 2016;164:245–57.

[41] Al Abri RS, El-Saadany EF, Atwa YM. Optimal placement and sizing method to improve the voltage stability margin in a distribution system using distributed generation. IEEE Trans. Power Syst. 2013;28(1):326–34. https://doi.org/10.1109/TPWRS.2012.2200049.

[42] A. L. A. Syrri and P. Mancarella, "Reliability evaluation of demand response to increase distribution network utilisation," *2014 Int. Conf. Probabilistic Methods Appl. to Power Syst. PMAPS 2014 - Conf. Proc.*, 2014, doi: 10.1109/PMAPS.2014.6960658.

[43] Swarnkar A, Gupta N, Niazi KR. Adapted ant colony optimization for efficient reconfiguration of balanced and unbalanced distribution systems for loss minimization. Swarm Evol. Comput. Sep. 2011;1(3):129–37. https://doi.org/10.1016/j.swevo.2011.05.004.

[44] Gholami K, Karimi S, Rastgou A. Fuzzy risk-based framework for scheduling of energy storage systems in photovoltaic-rich networks. J. Energy Storage Aug. 2022;52:104902. https://doi.org/10.1016/j.est.2022.104902.

[45] Niknam T, Kavousifard A, Aghaei J. Scenario-based multiobjective distribution feeder reconfiguration considering wind power using adaptive modified particle swarm optimisation. IET Renew. Power Gener. 2012;6(4):236–47. https://doi.org/10.1049/iet-rpg.2011.0256.

[46] Gholami K, Karimi S, Anvari-Moghaddam A. Multi-objective stochastic planning of electric vehicle charging stations in unbalanced distribution networks supported by smart photovoltaic inverters. Sustain. Cities Soc. 2022. https://doi.org/10.1016/j.scs.2022.104029.